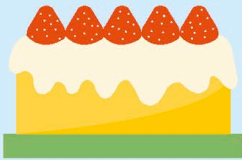


Decomposition in action

Decomposition is a lot like baking a cake. Both involve a task and some tools. In baking, the task is to make a cake, and the tools are the bowl, spoon, oven, and ingredients. In computing, the task might be to write a program, and the tools are a computer and programming language. A good way to start is to look at the problem in more detail and break it down into smaller tasks.



△ Getting it right

Breaking down the steps and then successfully completing each one will result in getting the cake right. In computer science, it's important to know what the objective is before beginning to write code.



△ Getting it wrong

Not working in a step-by-step manner to bake a cake or build a solution will result in failure.

Important parts

Breaking down a problem into smaller parts helps find an effective solution. Each part must then be completed successfully and in the right order to get the required result.



Ingredients and preparation

To bake a cake, the first step is to buy or gather the ingredients. The right amount of each ingredient must then be prepared while the oven heats up to the correct temperature.



Timing and combining

Each ingredient must be added and combined at the right time before the mixture is put in the oven.



Baking

The next step is to ensure that the mixture is baked at the right temperature and for the required amount of time.



Finishing

Finally, the cake must be removed from the tray and allowed to cool. It can then be decorated so that it both looks and tastes nice.

REAL WORLD

Building a spaceship

No matter how complex a computer program is, it's made up of solutions to lots of tiny problems. The process of building a complicated model of a spaceship out of building blocks is similar. Each part is the solution to a sub-problem and combining the components creates the spaceship.



We often solve a **problem** by **breaking** it down into **smaller parts**.